

A Federated Learning Approach for Distributed Alzheimer’s Disease Diagnosis Using Brain MRI

João Vítor de Freitas Scarlatelli, Vitória Símil Araújo, Nagib Alexandre Verly Borjaili,

Vitor Dias Britto Militão, Yasmin Cassemiro Viegas, Henrique Cota de Freitas, Felipe Domingos da Cunha

Department of Computer Science, Pontifical Catholic University of Minas Gerais, Belo Horizonte, Brazil

Emails: {joao.scarlatelli, vitoria.araujo.1321449, nagib.alexandre, vdbmilitao, yasmin.cassemiro}@sga.pucminas.br, {cota, felipe}@pucminas.br

Abstract—This work presents the design, implementation, and evaluation of an automated classification approach for brain magnetic resonance images to support the diagnosis of Alzheimer’s disease, emphasizing the use of Federated Learning (FL) to preserve data privacy in distributed clinical environments. The proposed approach integrates Deep Learning, distributed and parallel computing within a federated setting orchestrated by the Flower framework. A ResNet-18 architecture pretrained on ImageNet is employed as the backbone model, enabling robust feature extraction from neuroimaging data. In this framework, only model parameters are shared between clients and the central server, ensuring that sensitive patient data remains locally stored at each institution. Our approach is implemented using PyTorch, allowing efficient task distribution and parallelization across multiple clients. Experimental results, obtained by training and validating the model in a simulated multi-client federated setup, demonstrate the feasibility of the proposed proof-of-concept as a privacy-aware approach toward multi-institutional clinical decision-support applications.

Index Terms—Alzheimer’s Disease, Convolutional Neural Networks, Distributed and Parallel Computing, Federated Learning, Magnetic Resonance Imaging, Medical Image Classification

I. INTRODUCTION

Alzheimer’s disease is one of the major public health challenges of the 21st century, characterized by a progressive neurodegenerative process that compromises essential cognitive functions, including memory, logical reasoning, spatial orientation, and functional capacity [1]. Early diagnosis is widely recognized as a key factor in delaying progression, enabling timely clinical interventions and directly improving the quality of life of affected individuals [2]. However, identifying the early stages of the disease remains a complex task, especially because it relies on specialized analysis of neuroimaging exams, such as structural magnetic resonance imaging (MRI), whose interpretation requires substantial expertise and considerable time [3].

Modern cities concentrate not only population and infrastructure, but also the critical health challenges that accompany urbanization, particularly the management of chronic and neurodegenerative diseases affecting aging urban populations [1]. Urban computing, broadly defined as the acquisition, integration, and analysis of heterogeneous data generated by diverse sources within urban spaces to address pressing societal problems [4], has increasingly expanded its scope to encompass smart healthcare as a core urban service domain [5].

In this context, neuroimaging-based diagnosis of conditions such as Alzheimer’s disease represents a paradigmatic urban computing challenge: large volumes of sensitive clinical data are produced across geographically distributed hospitals and diagnostic centers within and across cities, yet regulatory constraints and patient privacy requirements prevent their centralized aggregation [6].

The heterogeneous and distributed nature of urban clinical networks calls for computational approaches that enable collaborative intelligence without compromising data locality. Approaches based on Artificial Intelligence and Deep Learning have shown promise by automating medical image analysis tasks, reducing the clinical workload, and increasing the potential to detect subtle patterns that might go unnoticed in manual evaluations [7]. Among these approaches, Convolutional Neural Networks (CNNs) stand out for their ability to learn hierarchical representations of complex data, making them particularly suitable for brain image classification tasks. By employing pre-trained models, such as ResNet-18, previously learned features can be adapted to a new domain, even when the available dataset is limited [8].

Despite the technical viability of Deep Learning, the consolidation of massive medical datasets poses severe ethical, legal, and security challenges. Neuroimaging data, particularly structural MRIs, constitute highly sensitive Protected Health Information (PHI). Beyond standard clinical metadata, high-resolution brain scans contain unique anatomical markers that could be reverse-engineered to identify patients, making centralized repositories highly susceptible to severe privacy breaches [6]. Furthermore, strict data protection regulations worldwide, such as the General Data Protection Regulation (GDPR) in Europe, the Health Insurance Portability and Accountability Act (HIPAA) in the United States, and the General Data Protection Law, from Portuguese *Lei Geral de Proteção de Dados* (LGPD) in Brazil, impose rigid constraints on the sharing and centralized storage of clinical records [9].

To address this critical data-sharing bottleneck, Federated Learning (FL) has emerged as an efficient and socially responsible solution, enabling the joint training of distributed models without transferring sensitive data across medical institutions. By keeping the raw MRI scans localized at the origin institution and sharing only the model parameters (weights and gradients) with a central server, this paradigm ensures greater privacy,

scalability, and adherence to data protection regulations, while simultaneously allowing the global model to benefit from the diversity of different patient bases [10].

In alignment with the Sustainable Development Goals (SDGs), particularly SDG 3, this work emphasizes promoting health and well-being by proposing an approach that facilitates more efficient and accessible diagnosis of Alzheimer's disease while respecting patient privacy. The present study aims to develop, train, and evaluate a deep neural network-based model for the automatic classification of the different levels of Alzheimer's disease present in the dataset used. The proposed modeling integrates preprocessing techniques, transfer learning, warm-up, and fine-tuning strategies, and supports distributed execution via the Flower library [11], thereby forming a controlled proof-of-concept toward scalable and privacy-aware clinical decision-support systems.

The main contributions of this work are summarized as follows:

- An end-to-end federated learning pipeline for Alzheimer's disease classification from brain MRI images, integrating transfer learning with a warm-up and fine-tuning training strategy within a privacy-preserving distributed architecture.
- An empirical evaluation of computational efficiency through parallelization experiments, analyzing the combined effect of GPU acceleration, mini-batching, and multi-worker data loading on training – an aspect largely overlooked in prior federated learning studies applied to medical imaging.
- An evaluation of the proposed approach under a controlled simulated multi-client federated setup with homogeneous data partitioning, comparing federated and centralized training modalities across standard classification metrics.
- A functional graphical user interface designed for clinical use, enabling healthcare professionals to interact with a classification system without requiring knowledge of the underlying computational infrastructure.

Unlike prior work that treats federated learning and model architecture in isolation, the proposed approach distinguishes itself by jointly addressing privacy preservation, classification performance, and client-side computational efficiency within a unified, reproducible pipeline.

This paper is organized as follows. Section II presents the related work on Alzheimer's disease classification using deep learning and federated learning approaches. Section III describes the proposed methodology, including the MRI classification pipeline, the federated learning framework, and the parallel computing optimization strategies. Section IV presents the experimental results, including centralized model performance, federated learning evaluation, and parallel computing analysis. Section V concludes the paper and discusses future research directions. Finally, we present our acknowledgment in the last section.

II. RELATED WORK

Deep learning-based approaches have been widely explored for Alzheimer's disease classification using structural MRI data. Early studies predominantly relied on centralized training pipelines and transfer learning with architectures such as VGG-16, VGG-19, ResNet, DenseNet, EfficientNet, and Inception-based models [3]. VGG-based models, although effective due to their straightforward convolutional architecture, typically require many parameters, leading to high computational cost and an increased risk of overfitting on limited medical datasets. DenseNet architectures improve feature reuse through dense connectivity, but their memory consumption can become a bottleneck, particularly in distributed or resource-constrained environments. More recent ensemble approaches combining multiple CNN architectures have reported competitive classification performance. However, they rely on centralized data aggregation, which introduces additional inference complexity [12].

A closely related work [13] investigates federated learning for Alzheimer's classification using T1-weighted MRI scans. Sahid et al. consider both IID (Independent and Identically Distributed) and non-IID data distributions across clients and demonstrate that federated learning enables privacy-preserving collaborative training. However, the reported maximum accuracy of approximately 84.75% remains notably lower than typical centralized approaches, highlighting challenges such as data heterogeneity, limited local datasets, and suboptimal convergence of the global model.

Recent works have further explored federated learning in medical imaging scenarios. Li et al. [14] proposed a privacy-preserving federated framework for medical image analysis, demonstrating the feasibility of collaborative training across institutions without sharing raw data. Although effective, such approaches often rely on deep, computationally expensive architectures, increasing both communication overhead and local training costs during federated rounds. Similarly, EfficientNet-based models [15] have been proposed to improve accuracy-efficiency trade-offs through compound scaling strategies, but they require careful tuning to maintain stability across heterogeneous client environments.

Despite these advances, most federated learning studies in medical imaging focus primarily on model architecture, aggregation strategies, and data distribution, while largely overlooking the optimization of local training performance. In particular, the use of hardware-aware optimizations, such as GPU acceleration, asynchronous data loading, and batch-level parallelism, is rarely discussed or systematically evaluated. This gap results in solutions that, although theoretically viable, may entail long training times and limited practicality in real-world clinical settings, where client-side computational efficiency is essential.

In contrast to prior work, this study proposes a unified approach that integrates a lightweight ResNet-18 architecture with transfer learning, warm-up and fine-tuning strategies, federated training using the Flower framework, and explicit

parallel computing optimizations. The selection of ResNet-18 provides a favorable balance between representational capacity and computational efficiency, making it suitable for resource-constrained clients. Furthermore, incorporating GPU acceleration, batching, and multi-worker data loading directly addresses the underexplored challenge of client-side efficiency in federated learning. By jointly considering privacy preservation, model performance, and computational scalability, the proposed approach advances beyond existing solutions and moves closer to practical deployment in multi-institutional clinical settings.

III. METHODOLOGY

The proposed methodology integrates deep learning, federated learning, and parallel computing to develop a scalable, privacy-preserving system for the automated classification of brain magnetic resonance imaging (MRI) scans. The overall process is structured into three complementary layers: (i) image classification using a convolutional neural network, (ii) distributed training through federated learning, and (iii) parallel computing optimization to improve computational efficiency.

This integrated pipeline enables each client to process data locally, train the model in multiple stages, and share only the model parameters with a central server, thereby ensuring compliance with data privacy requirements. Additionally, hardware acceleration and parallel data processing enable efficient training even in large-scale, multi-client environments.

A. MRI Image Classification

MRI image classification is performed using a deep learning pipeline implemented in PyTorch and based on the ResNet-18 architecture, pretrained on ImageNet [16], [17]. The use of a lightweight residual architecture enables efficient feature extraction while balancing performance and computational cost, which is essential for both centralized and federated scenarios.

The dataset used in this study is the Augmented Alzheimer MRI Dataset, publicly available on Kaggle [18]. It consists of labeled brain MRI images categorized into four classes: MildDemented, ModerateDemented, NonDemented, and VeryMildDemented. The dataset includes both original and augmented samples, providing greater variability and supporting the training of deep learning models with improved generalization.

The dataset is organized using the ImageFolder API into three subsets: training, validation, and testing, following a target proportion of 70%/15%/15%. Due to integer truncation during dataset splitting, the exact number of samples per class may vary slightly, with the remaining samples allocated to the test set. The class distribution is preserved across all subsets to ensure consistency during model evaluation.

Preprocessing includes resizing all images to 224×224 pixels, converting them to tensor format, and normalizing them using ImageNet statistics. These steps ensure compatibility with the pretrained feature distribution and stabilize the training process. In the training set, lightweight data augmentation techniques are applied, including random rotations (up to 10 degrees)

and horizontal flipping, increasing variability while preserving relevant anatomical structures.

The training process is divided into two stages to improve convergence and reduce overfitting. In the first stage (warm-up), all convolutional layers are frozen, and only the final fully connected layer is trained using the Adam optimizer [19] with a learning rate of 1×10^{-4} and the Cross-Entropy Loss function (`nn.CrossEntropyLoss`). This stage adapts the classifier to the new domain while preserving pretrained feature representations.

In the second stage (fine-tuning), deeper layers of the network (specifically *layer3*, *layer4*, and the final classification layer) are unfrozen, and training continues using a reduced learning rate of 1×10^{-5} . This allows the model to refine features related to cortical morphology and structural brain changes associated with Alzheimer’s disease progression.

Training is performed using mini-batches of size 32. This configuration was selected based on convergence quality rather than maximum parallelism throughput: smaller batches produce more frequent gradient updates, which improves convergence in transfer learning scenarios with limited domain-specific data [20]. The model optimization process includes epoch-wise monitoring and automatic checkpointing to preserve the best-performing model configuration.

Model evaluation is performed on the test set using standard classification metrics, including accuracy, precision, recall, F1-score, and specificity. Macro-averaging is used to balance evaluation across all diagnostic classes, mitigating class imbalance and ensuring fair performance assessment. Additionally, a confusion matrix is generated from the test predictions and used as an analytical tool to examine classification errors and identify misclassification patterns, particularly among clinically similar stages of Alzheimer’s disease. The detailed results and interpretation of this matrix are presented and discussed in the Results section.

B. Distributed Computing with Federated Learning

The distributed training layer is implemented using the Flower framework [11], enabling a federated learning environment in which multiple clients collaboratively train a global model without sharing raw data. This approach is particularly suitable for medical imaging applications, where data privacy and regulatory constraints limit centralized data aggregation.

The federated learning process follows a client-server architecture, in which a central server coordinates training, and multiple clients perform local model updates. Initially, the server distributes the current global model parameters to a subset of participating clients. Each client then trains the model locally using its private dataset, following the same two-stage training strategy described in Section III-A, including warm-up and fine-tuning phases.

After local training, clients send only the updated model parameters and aggregated metrics back to the server. The server then applies the Federated Averaging (FedAvg) algorithm [21] to combine the client updates and produce a new global model. The aggregation process is defined as:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_t^k \quad (1)$$

where w_t^k represents the model parameters trained by client k , n_k is the number of samples in client k , and $n = \sum_{k=1}^K n_k$ is the total number of samples across all clients. This weighted aggregation ensures that clients with larger datasets contribute proportionally to the global model update.

The training proceeds iteratively across multiple communication rounds, with the updated global model redistributed to clients at each round. This iterative refinement allows the global model to learn from distributed data sources while preserving data locality and minimizing communication overhead.

In the implemented environment, all available clients participate in each training round (client fraction equal to 1.0), ensuring full utilization of the distributed dataset. A minimum of three clients is maintained per round to ensure statistical robustness and to reduce bias arising from individual data distributions.

To enhance monitoring and analysis of the federated training process, the aggregation strategy is extended to record additional metrics during training. Specifically, the `aggregate_fit` function computes the average training time, the mean local accuracy, and the dataset size across clients, whereas the `aggregate_evaluate` function aggregates the validation metrics reported by each client. This enables tracking of global loss, global accuracy, round duration, and variability across clients.

The federated learning setup assumes a homogeneous data distribution across clients for experimental consistency, although the framework supports heterogeneous (non-IID) scenarios. This design choice facilitates controlled evaluation of the federated training process while maintaining alignment with realistic multi-institutional deployment conditions.

Overall, the federated learning layer enables collaborative model training while preserving data privacy, reducing data transfer requirements, and supporting scalable deployment across multiple institutions.

C. Parallel Computing Optimization

The Parallel Computing implementation is based on PyTorch's native support for parallel tensor operations and GPU acceleration. Training is executed on an NVIDIA GPU using CUDA, enabling automatic offloading of computationally intensive operations through the `.to(device)` method. This allows matrix operations and gradient computations to be performed in parallel across thousands of CUDA cores, significantly reducing training time.

In addition to hardware-level parallelism, data loading is optimized using multiple worker processes in the `DataLoader`. This approach enables asynchronous loading and preprocessing of image batches, thereby minimizing input/output (I/O) bottlenecks and ensuring efficient GPU utilization during training. By overlapping data preparation and model execution, the pipeline achieves higher throughput and improved resource utilization.

Batch processing (mini-batching) is also employed to maximize parallel execution. Multiple images are processed simultaneously in both the forward and backward passes, improving computational efficiency and stabilizing gradient updates. The batch size is configured to balance memory usage and computational performance, ensuring efficient utilization of GPU resources.

The parallelization strategy operates at three complementary levels: (i) hardware-level parallelism via GPU and CUDA acceleration, (ii) data-level parallelism through multi-worker data loading, and (iii) batch-level parallelism via mini-batch training. These mechanisms collectively reduce training latency and enable scalable experimentation.

This design is particularly relevant in federated learning scenarios, where local training efficiency directly impacts the duration of each communication round. By optimizing client-side computation, the proposed approach reduces training overhead and supports practical deployment in real-world clinical environments with limited computational resources.

D. Graphical User Interface

The suggested computational interface developed for the brain magnetic resonance imaging-based Alzheimer's disease classification system was implemented using PyQt5, a Python library for building graphical user interfaces with the Qt Toolkit. The interface was designed to seamlessly integrate with the trained ResNet-18 model, enabling healthcare professionals to interact with the classification system without requiring knowledge of the underlying computational infrastructure.

The application¹ follows a modular architecture organized around three main structural components. The navigation sidebar provides a lateral menu that the user uses to access the CNN Classification module. The main content area occupies the central region of the window and dynamically loads the appropriate widget according to the active navigation selection. The specialized widget system comprises reusable components responsible for distinct classification sub-tasks, promoting separation of concerns and facilitating future extensibility.

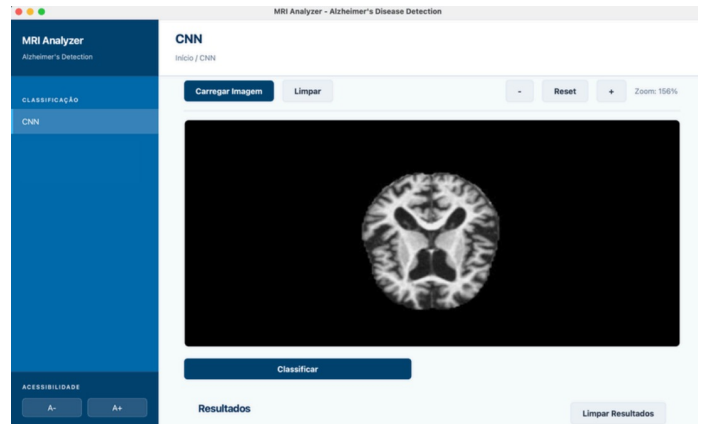


Fig. 1. Layout of the CNN Classification Module.

¹<https://github.com/NagibAlexandre/AlzheimerMRI>

Classification is performed in real time by submitting the preprocessed image directly to the deep learning model. The interface then presents the output as a set of per-class probabilities corresponding to the four diagnostic categories: MildDemented, ModerateDemented, NonDemented, and VeryMildDemented. This probabilistic presentation provides clinicians with a nuanced view of the model’s confidence distribution rather than a single categorical prediction, supporting more informed decision-making.

IV. RESULTS

This section presents the experimental results obtained from the proposed approach. The analysis is organized into three complementary perspectives: centralized training, federated learning, and parallel computing performance. The centralized experiment evaluates the predictive capacity of the ResNet-18 model when trained on the complete training subset and evaluated on the independent test subset. The federated experiment analyzes the behavior of the global model when training is distributed across clients using the Flower framework and FedAvg aggregation. Finally, the parallel performance evaluation investigates the impact of batch size and the number of data-loading workers on execution time and scalability.

A. Centralized Model Results

The centralized model was trained using the full training subset and evaluated on the independent test subset. After training, the model generated class predictions for all test images, which were compared against the true labels to compute the confusion matrix and the classification metrics.

The centralized CNN achieved high performance on the MRI classification task (Table I), achieving an accuracy of 97.38%, a precision of 97.66%, a recall of 97.74%, and an F1-score of 97.70%. The model also achieved a macro specificity of 99.07%, indicating a low false-positive rate. These results suggest that the ResNet-18 model extracted discriminative visual patterns from the MRI images and effectively distinguished among the four diagnostic classes.

TABLE I
CENTRALIZED MODEL CLASSIFICATION RESULTS

Metric	Value
Accuracy	97.38%
Precision (macro)	97.66%
Recall (macro)	97.74%
F1-score (macro)	97.70%
Specificity (macro)	99.07%
False Positive Rate	0.93%
False Negative Rate	2.26%
Training time	≈ 54 min

As shown in Fig. 2, the classification errors are mainly concentrated among clinically closer classes. This behavior is expected in Alzheimer’s disease classification, since adjacent stages may present subtle visual differences in structural MRI scans [3]. Therefore, the confusion matrix complements the numerical metrics by showing where the model is more likely to confuse disease severity levels.

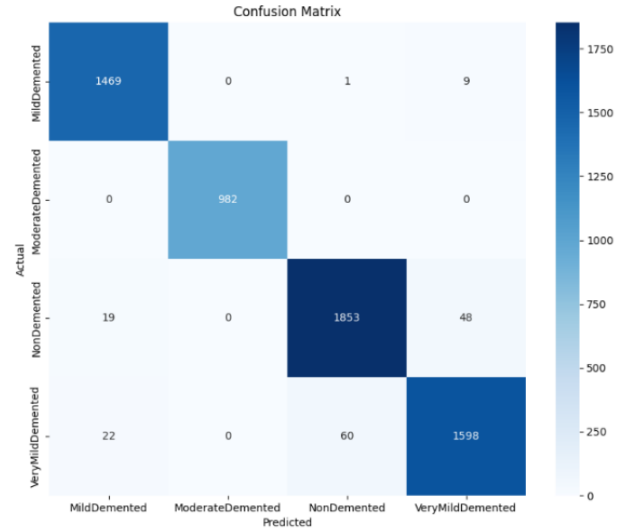


Fig. 2. Confusion matrix generated by the centralized CNN model

Although the centralized approach achieved the best predictive performance, it required approximately 54 minutes to complete training. This computational cost motivated the adoption of distributed and parallel strategies to reduce training time, improve scalability, and support scenarios in which data cannot be centralized due to privacy restrictions.

B. Federated Learning Results

The federated learning experiment was conducted using the Flower framework with a client-server architecture. In this setup, each client trained the ResNet-18 model locally using its own subset of MRI images, while the central server coordinated the aggregation of model parameters through the Federated Averaging algorithm. Raw MRI images were not shared between clients and the server; only updated model weights and aggregated metrics were transmitted.

The federated setup used three clients. Each client performed local training and returned its updated parameters to the server, which then aggregated the received updates to produce the next global model. The evaluation considered both the local metrics obtained by each client and the final behavior of the global federated model on the test set. This distinction is important because local client metrics² reflect the performance observed during distributed training, while the confusion matrix represents the final evaluation of the aggregated global model.

The results in Table II show that the federated training process achieved consistent behavior across the three clients. All clients obtained the same accuracy of 90.83%, precision of 91.71%, recall of 92.05%, and specificity of 96.68%. This similarity suggests that the local evaluation behavior was consistent across clients, possibly due to a relatively balanced data partition and stable FedAvg aggregation.

²Client metrics represent mean training accuracy across warmup and fine-tuning epochs (in-sample); they are not directly comparable to the global model test-set evaluation in Table III.

TABLE II
FEDERATED LEARNING RESULTS ACROSS CLIENTS

Metric	Client 1	Client 2	Client 3
Accuracy	90.83%	90.83%	90.83%
Precision	91.71%	91.71%	91.71%
Recall	92.05%	92.05%	92.05%
Specificity	96.68%	96.68%	96.68%

It is important to note that the client-level local metrics differ structurally from the final test-set performance of the global federated model. The local accuracy of 90.83% corresponds to the mean training accuracy across warmup and fine-tuning epochs, computed over each client’s own partition during the fit phase, reflecting in-sample performance rather than generalization. In contrast, the global model accuracy of 86.95% (Table III) is evaluated on the held-out test set via the aggregated FedAvg model. The consistent equality of metrics across all three clients is expected under homogeneous data partitioning and full client participation per round, confirming the correctness of the federated pipeline rather than indicating a limitation.

The macro-averaged metrics (Table III) were 88.20% precision, 88.61% recall, 88.37% F1-score, and 95.46% specificity. These values indicate that although the federated model performed worse than the centralized CNN, it still maintained competitive classification performance while preserving the distributed nature of the training process.

TABLE III
FEDERATED MODEL TEST-SET RESULTS

Metric	Value
Accuracy	86.95%
Precision (macro)	88.20%
Recall (macro)	88.61%
F1-score (macro)	88.37%
Specificity (macro)	95.46%
Execution time	594 s ($\approx$ 10 min)

The total federated training time across the three clients was 594 seconds, approximately 10 minutes. Compared with the centralized model, which required approximately 54 minutes, the federated setup distributed the computational workload across clients while preserving the privacy of local MRI data. However, the federated model achieved lower predictive performance than the centralized CNN, as expected, because it was trained on local subsets rather than a single centralized dataset.

The final global federated model was also evaluated through the confusion matrix shown in Fig. 3. This matrix was generated by applying the aggregated federated model to the test set and comparing the predicted labels with the true labels of each MRI image. Therefore, it provides a class-level analysis of the final federated model, allowing the identification of the most frequent misclassification patterns.

Table IV details the federated training experiment that was repeated across five independent runs ($n = 5$), comprising four automated runs and one manually conducted run under

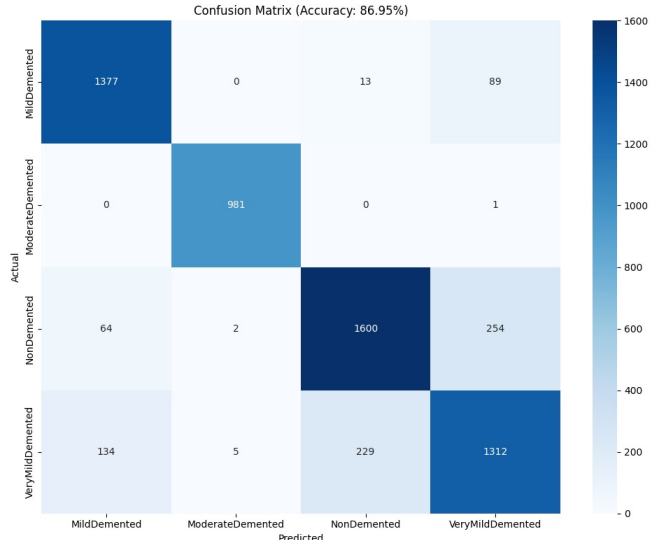


Fig. 3. Confusion matrix generated by the federated model.

identical model configuration. The global model accuracy, evaluated on the validation set after FedAvg aggregation, yielded a mean of 89.55%, a standard deviation of 1.47%, and a 95% confidence interval of [87.72%, 91.38%] (using a t -distribution with $df = 4$). The observed variance across runs reflects the inherent stochasticity of the training process, introduced by data augmentation and mini-batch shuffling, as well as natural variability between execution environments. These results indicate that the federated pipeline produces consistent and reproducible outcomes under the controlled homogeneous setup, supporting the validity of the reported metrics.

TABLE IV
FEDERATED MODEL VALIDATION-SET RESULTS.

Metric	Value	95% CI
Mean Accuracy	89.55%	[87.72%, 91.38%]
Std. Deviation	1.47%	—
$n = 5$ runs, t -distribution, $df = 4$, $t_{crit} = 2.7764$		

The best class-level performance was observed for the ModerateDemented and MildDemented classes, with 981 and 1377 correct predictions, respectively. Most misclassifications were concentrated between the NonDemented and VeryMildDemented classes, which are clinically adjacent and present subtle structural differences in MRI scans [3].

Table V summarizes the overall comparison between the centralized and federated models. The centralized CNN achieved 97.38% accuracy (Table I), while the final aggregated federated model reached 86.95% accuracy on the test set (Table III), reflecting the expected trade-off between data locality and predictive performance.

Overall, the federated results highlight the trade-off between predictive performance and privacy preservation. Although centralized training achieved higher accuracy, federated learning enabled collaborative model training without centralizing

TABLE V
COMPARISON BETWEEN CENTRALIZED AND FEDERATED MODELS.

Metric	Centralized CNN	Federated Model
Accuracy	97.38%	86.95%
Precision	97.66%	88.20%
Recall	97.74%	88.61%
F1-score	97.70%	88.37%
Specificity	99.07%	95.46%
Execution time	≈ 54 min	≈ 10 min

sensitive medical images. This is relevant in clinical scenarios, where privacy restrictions may prevent direct data sharing between institutions.

C. Parallel Computing Results

In addition to predictive performance, the computational efficiency of the proposed approach was evaluated through parallelization experiments. These experiments analyzed the impact of batch size and the number of data loading workers on execution time and speedup. The parallel strategy combined GPU acceleration, batch-level processing, and multi-worker data loading. The experiments were executed using an NVIDIA RTX 3070 Ti GPU and a Ryzen 7 9800X3D processor.

Table VI summarizes the impact of the number of data loading workers on execution time for different batch sizes. A consistent improvement in speedup is observed as the number of workers increases from 2 to 4 across all configurations, indicating more efficient data loading and better utilization of computational resources. However, increasing the number of workers to 8 does not result in higher speedups. This suggests the presence of system overhead, synchronization cost, and I/O contention.

TABLE VI
ABSOLUTE SPEEDUP ACHIEVED FOR DIFFERENT BATCH SIZES AND NUMBER OF WORKERS, USING SEQUENTIAL EXECUTION TIME $T_{seq} = 1324.71$ S AS BASELINE

Batch Size	2 Workers	4 Workers	8 Workers
32	4.77×	10.85×	10.02×
64	9.13×	12.31×	10.89×
128	8.00×	12.67×	11.50×

The results obtained with the sequential baseline of 1324.71 seconds indicate a substantial reduction in execution time when parallelism is introduced. For a batch size of 32, the speedup increased from 4.77× with 2 workers to 10.85× with 4 workers, then slightly decreased to 10.02× with 8 workers. For batch sizes of 64 and 128, the highest speedups were achieved with 4 workers, reaching 12.31× and 12.67×, respectively.

These results indicate that moderate parallelism strikes the best balance between data-loading efficiency and system overhead. Although increasing the number of workers improves throughput up to a certain point, excessive parallelism may introduce contention and reduce performance gains. Therefore, the 4-worker configuration is the most efficient among the tested configurations.

Overall, the parallel computing experiments demonstrate that GPU acceleration can substantially reduce training time. The

optimized configurations reduced execution time from more than 22 minutes in the sequential baseline to approximately 1.5–2 minutes. This result is especially important in federated learning scenarios, where local client training time directly affects the duration of each communication round.

V. CONCLUSION

This study successfully demonstrated the viability and impact of integrating Deep Learning, Federated Learning, and Parallel Computing to address the complex challenges of early Alzheimer’s disease diagnosis. By utilizing a pre-trained ResNet-18 architecture within a federated environment, the proposed system effectively extracts hierarchical features from structural MRI scans to classify the progressive stages of dementia. The federated approach mitigates the severe data-sharing bottlenecks inherent in medical research. By keeping raw, sensitive Protected Health Information (PHI) strictly localized and sharing only model parameters, the framework supports compliance with stringent data protection regulations such as GDPR, HIPAA, and Brazilian LGPD, while still enabling collaborative learning across simulated distributed data partitions.

A significant contribution of this work lies in the empirical analysis of the system’s computational efficiency. The parallelization strategy, which includes GPU acceleration, multi-worker data loading, and mini-batching, substantially reduced training latency compared to a sequential baseline of 1324.71 seconds. The performance evaluation revealed that a moderate level of parallelism is optimal; specifically, using 4 data-loading workers consistently yielded the highest throughput, achieving a peak absolute speedup of 12.67x with a batch size of 128. The data clearly indicates that while parallelism yields substantial performance gains, exceeding optimal worker thresholds (e.g., 8 workers) introduces synchronization and I/O contention overhead, leading to diminishing returns.

Ultimately, this integrated architecture provides a proof-of-concept toward robust, scalable, and socially responsible clinical decision-support systems that align with global health objectives (SDG 3). By framing distributed clinical AI within an urban computing perspective, the proposed approach contributes to the broader vision of smart healthcare infrastructure, where geographically distributed institutions collaborate without compromising patient data locality [4]. It enables high-throughput, collaborative model training across clinical environments with varying computational resources. However, it remains paramount to emphasize that this computational system is designed as a supportive tool to reduce clinical workload and highlight subtle neurodegenerative patterns. It does not replace the critical evaluation of qualified healthcare professionals, whose expertise remains essential for the final clinical interpretation and validation of the automated results.

As a natural progression of this research, several directions warrant further investigation. First, the federated setup should be extended to include a larger number of clients and heterogeneous (non-IID) data-partitioning strategies, more closely reflecting the variability observed in real urban clinical

networks, where institutions differ in patient demographics, imaging protocols, and dataset sizes [4], [5]. Second, advanced aggregation algorithms such as FedProx [22] and SCAFFOLD [23] should be evaluated as alternatives to FedAvg, particularly under heterogeneous conditions where client drift may compromise global model convergence. Third, comparisons with alternative convolutional architectures, such as EfficientNet and DenseNet, would provide a more comprehensive assessment of the accuracy-efficiency trade-off in federated medical imaging scenarios [15]. Fourth, future experiments should extend the reproducibility analysis to heterogeneous setups, reporting variance and confidence intervals under non-IID data partitioning conditions, where training stochasticity is expected to be significantly higher. Finally, incorporating explainable AI (XAI) techniques, such as gradient-weighted class activation mapping (Grad-CAM), would enhance model transparency, enabling clinicians to visually verify the anatomical markers that drive the network's classification decisions.

ACKNOWLEDGMENT

The authors would like to thank the Brazilian National Council for Scientific and Technological Development (CNPq; grant numbers 311697/2022-4 and 402837/2024-0), the Research Support Foundation of the State of Minas Gerais (FAPEMIG; grant number APQ-05058-23), and PUC Minas (FIP grant number 2026/35258). This project was also supported by the Brazilian Ministry of Science, Technology, and Innovation, with resources from Law n° 8,248, of October 23, 1991, within the scope of PPI-SOFTEX, coordinated by Softex and published Arquitetura Cognitiva (Phase 3), DOU 01245.003479/2024-10.

REFERENCES

- [1] Alzheimer's Disease International, "World alzheimer report 2021," 2021, accessed: April 25, 2026. [Online]. Available: <https://www.alzint.org/resource/world-alzheimer-report-2021/>
- [2] A. Porsteinsson, R. Isaacson, S. Knox, M. Sabbagh, and I. Rubino, "Diagnosis of early alzheimer's disease: Clinical practice in 2021," *The Journal of Prevention of Alzheimer's Disease*, vol. 8, no. 3, pp. 371–386, 2021. [Online]. Available: <https://doi.org/10.14283/jpad.2021.23>
- [3] X. Sun, J. Zhu, R. Li, Y. Peng, and L. Gong, "The global research of magnetic resonance imaging in alzheimer's disease: a bibliometric analysis from 2004 to 2023," *Frontiers in Neurology*, vol. Volume 15 - 2024, 2025. [Online]. Available: <https://doi.org/10.3389/fneur.2024.1510522>
- [4] Y. Zheng, L. Capra, O. Wolfson, and H. Yang, "Urban computing: Concepts, methodologies, and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 5, no. 3, Sep. 2014. [Online]. Available: <https://doi.org/10.1145/2629592>
- [5] S. Pandya, G. Srivastava, R. Jhaveri, M. R. Babu, S. Bhattacharya, P. K. R. Maddikunta, S. Mastorakis, M. J. Piran, and T. R. Gadekallu, "Federated learning for smart cities: A comprehensive survey," *Sustainable Energy Technologies and Assessments*, vol. 55, p. 102987, 2023. [Online]. Available: <https://doi.org/10.1016/j.seta.2022.102987>
- [6] A. K. Conduah, S. Ofoe, and D. Siaw-Marfo, "Data privacy in healthcare: Global challenges and solutions," *Digital Health*, 2025. [Online]. Available: <https://doi.org/10.1177/20552076251343959>
- [7] S. Dardouri, "An efficient method for early alzheimer's disease detection based on mri images using deep convolutional neural networks," *Frontiers in Artificial Intelligence*, vol. Volume 8 - 2025, 2025. [Online]. Available: <https://doi.org/10.3389/frai.2025.1563016>
- [8] S. B. Francis and J. Prakash Verma, "Deep cnn resnet-18 based model with attention and transfer learning for alzheimer's disease detection," *Frontiers in Neuroinformatics*, vol. Volume 18 - 2024, 2025. [Online]. Available: <https://doi.org/10.3389/fninf.2024.1507217>
- [9] C. Konnoth, "Ai and data protection law in health," 2024, accessed: April 25, 2026. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK613196/>
- [10] A. Arini, F. Fahrianto, and A. Ramadhan, "Implementation of federated learning for alzheimer's disease classification using fedadagrad algorithm," *IJID (International Journal on Informatics for Development)*, no. 2, p. 659–671, Jan. 2026. [Online]. Available: <https://ejournal.uin-suka.ac.id/saintek/ijid/article/view/5045>
- [11] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, J. Fernandez-Marques, Y. Gao, L. Sani, K. H. Li, T. Parcollet, P. P. B. d. Gusmao, and N. D. Lane, "Flower: A friendly federated learning research framework," *arXiv preprint arXiv:2007.14390*, 2020. [Online]. Available: <https://arxiv.org/pdf/2007.14390>
- [12] S. Fathi, A. Ahmadi, A. Dehnad, M. Almasi-Dooghaee, M. Sadegh, and Alzheimer's Disease Neuroimaging Initiative, "A deep learning-based ensemble method for early diagnosis of Alzheimer's disease using MRI images," *Neuroinformatics*, vol. 22, no. 1, pp. 89–105, 2024. [Online]. Available: <https://doi.org/10.1007/s12021-023-09646-2>
- [13] M. A. Sahid, M. P. Uddin, H. Saha, and M. R. Islam, "Towards privacy-preserving alzheimer's disease classification: Federated learning on t1-weighted magnetic resonance imaging data," *Digital Health*, vol. 10, 2024. [Online]. Available: <https://doi.org/10.1177/20552076241295577>
- [14] W. Li, F. Milletari, D. Xu, N. Rieke, J. Hancox, W. Zhu, M. Baust, Y. Cheng, S. Ourselin, M. J. Cardoso, and A. Feng, "Privacy-preserving federated brain tumour segmentation," in *Machine Learning in Medical Imaging*. Cham: Springer International Publishing, 2019, pp. 133–141. [Online]. Available: https://doi.org/10.1007/978-3-030-32692-0_16
- [15] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, K. Chaudhuri and R. Salakhutdinov, Eds., vol. 97. PMLR, 09–15 Jun 2019, pp. 6105–6114. [Online]. Available: <https://proceedings.mlr.press/v97/tan19a.html>
- [16] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. [Online]. Available: <https://doi.org/10.1109/CVPR.2016.90>
- [17] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255. [Online]. Available: <https://doi.org/10.1109/CVPR.2009.5206848>
- [18] U. Injo, "Augmented alzheimer mri dataset," <https://www.kaggle.com/datasets/uraninjo/augmented-alzheimer-mri-dataset>, 2023, accessed: 2026.
- [19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *International Conference on Learning Representations (ICLR)*, 2015. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [20] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, "On large-batch training for deep learning: Generalization gap and sharp minima," in *5th International Conference on Learning Representations (ICLR)*, 2017. [Online]. Available: <https://arxiv.org/abs/1609.04836>
- [21] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, A. Singh and J. Zhu, Eds., vol. 54. PMLR, 20–22 Apr 2017, pp. 1273–1282. [Online]. Available: <https://proceedings.mlr.press/v54/mcmahan17a.html>
- [22] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proceedings of Machine Learning and Systems*, vol. 2, 2020, pp. 429–450.
- [23] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, H. D. III and A. Singh, Eds., vol. 119. PMLR, 13–18 Jul 2020, pp. 5132–5143. [Online]. Available: <https://proceedings.mlr.press/v119/karimireddy20a.html>